In summary, essentially 100% of the Pu, 85−100% of the Nd, 75−90% of the Ru and 60−85% of the Tc were retained within the reactor zones. The migrating fission products were held within a few tens of meters of these zones. Thermodynamic calculations of the temperature dependent solubilities indicate that the loss of fissiogenic elements is diffusion controlled, whereas, retention in the surrounding rocks is due to temperature dependent deposition from an aqueous solution.

While the conditions at Oklo differ in a number of aspects from those expected in nuclear repository sites, they frequently were much less favorable to retention of the radionuclides. The lack of migration of the actinides and the much slower release of Tc agree with the predictions of laboratory studies and indicate their value in validating the safety of nuclear repositories.

22.10. PERFORMANCE ASSESSMENTS OF WASTE REPOSITORIES

In this Chapter we have presented some information on studies of the environmental behavior of some fission products and actinides. We have shown how laboratory data can be used in speciation calculations to predict solubilities, etc. The knowledge gained from studies of natural analog sites and, very importantly, from the Oklo natural reactor site, has been reviewed. All of these types of studies and data are of use in assessing the probable ability of nuclear waste repositories to protect the public over the time needed required for the radioactivity to decay. This time is largely dependent on used nuclear fuel handling as discussed previously and thsu the time required may vary between some thousends of years for a full transmutation cycle to more than 100 000 years for the once through cycle. The most common design goal for these repositories is that no radioactivity shall be released within the next 1 000 years, and that the leakage after that time would be so small that it presents no danger to living species. The Environmental Protection Agency (EPA), USA, standard requires that no more than 1 000 cancers shall be caused in 10 000 years by radionuclides released from the waste, using the ICRP predictions of radiation dose effects (Ch. 15).

In the case of direct disposal of used nuclear fuel, the initiated radioactive inventory consists of actinides, fission products and activation products. As noted previously, (Ch. 21) the shorter lived fission products, such as ^{90}Sr and ^{137}Cs, and transuranic elements, such as ^{238}Pu, ^{241}Pu, ^{244}Cm, are the main contributors to the radioactivity. However, performance assessments strongly indicate that the waste form matrix and the near field engineered barriers (e.g. clay backfill, etc.), can successfully retain and prevent any migration to the far field environment for one thousand years and probably much longer ($>10^4$ years). After the first thousand years the long lived nuclides such as ^{129}I, ^{135}Cs, ^{126}Sn, ^{99}Tc and ^{79}Se among the fission products, ^{226}Ra, and the actinides ^{234}U, ^{236}U, ^{237}Np, ^{239}Pu, ^{240}Pu, and ^{241}Am become the major concern.

22.10.1. Release scenarios

Two major types of scenarios are considered in the performance assessments for release of radionuclides from the repository. One scenario evaluates all the processes which are expected to occur normally in the region of the repository which could effect the rate of release and migration. In this scenario, ground water penetrates the waste packages and leaches the radionuclides which, then, can migrate through and out of the repository, Figure 22.12. R_1, R_2, etc represent mathematical models describing the migration of one particular radionuclide in that specific region. The R_i's are then added into the overall equation.

To assess the release by this scenario, it is necessary to evaluate the rate of release from the package, the flow rate of the underground fluids, the speciation and solubility of the different radionuclides and their diffusion (migration) rates.

The second scenario includes "disturbed" conditions which could be the result of geologic events such as earthquakes, volcanic activity and changes in hydrological conditions. This scenario also includes the effect of "human intrusion," in which people in future generations unknowingly penetrate a repository and release a portion of its radioactive contents to the earth's surface via the groundwater system. For example, this release could be the result of a drilling operation. Such events are assumed to occur over the history of the repository including some at very early times (less than 1 000 years). In this case, sorption and half-life and, for the most part, solubility are not major factors in determining the potential release. The dominant contribution would be from transuranic elements (mainly ^{239}Pu, ^{240}Pu, ^{241}Am) with some contributions from fission products.

22.10.2. Canister dissolution

The most common canister materials are copper and iron. In rocks, where the ground water is reducing (as e.g. in the Canadian and Scandinavian shields), copper is practically insoluble as shown by the existence of native copper, million years old, found in this environment. Detailed studies have shown that the radiation from the waste nuclides have a very small effect on the dissolution. It is therefore predicted that a 50 mm thick copper canister will be intact for at least one million years. Iron, or steel, can be expected to dissolve more rapidly, especially in oxidizing groundwater. This anoxic oxidation will produce hydrogen gas which will contribute to keep the general redox potential reducing which would be important in limiting the waste nuclide migration.

As the metal encapsulation is dissolved or cracks (Figure 22.12.B), the radionuclides in the waste matrix (UO_2 or glass) will be released mainly with the dissolution rate of the matrix (congruent dissolution). A reasonable figure for the glass dissolution rate is $2 \times 10^{-3}\ g\ m^{-2}\ d^{-1}$ exposed glass surface, assuming no limit with regard to the solubility

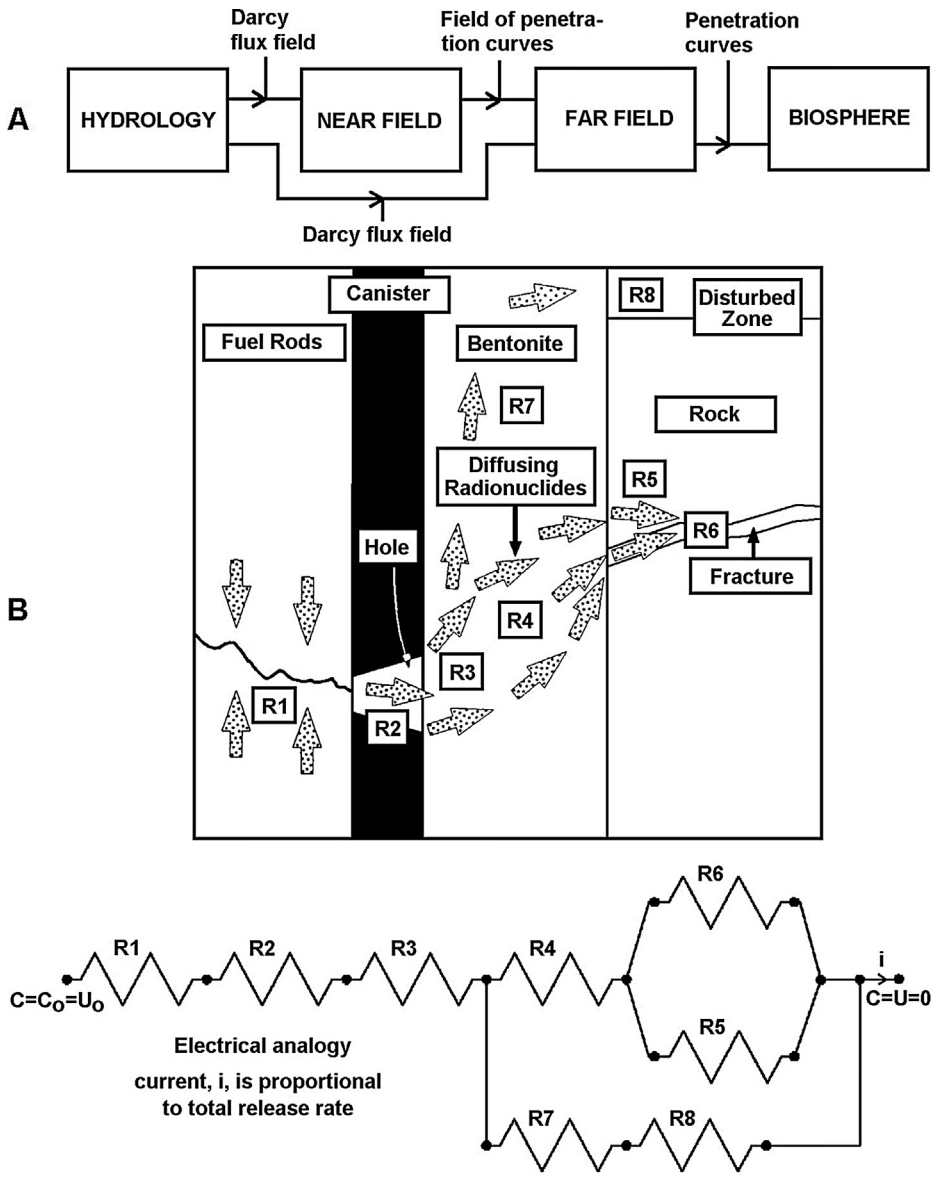


Figure 22.12 Coupling of models in model chain for safety assessment of leakage of radionuclides from fractured fuel rod and canister in a bentonite filled rock hole. (*From SKB91*).

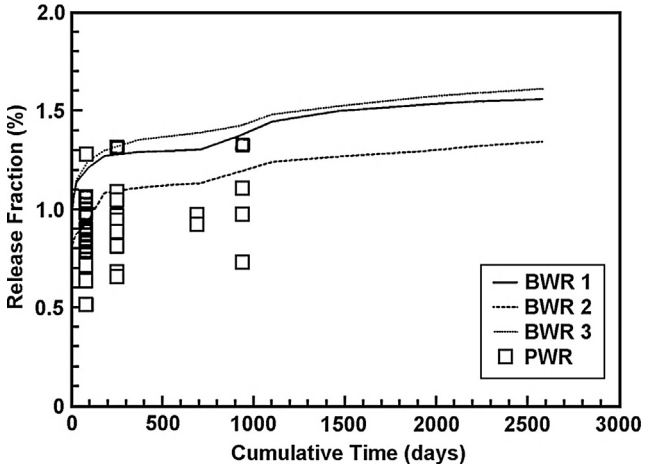


Figure 22.13 Fraction of released cesium from BWR and PWR fuels in different waters as a function of contact time (*From SKB91*).

product (i.e. unlimited amount of water); this corresponds to a corrosion rate of the glass surface of $\sim 2.7 \times 10^{-4}$ mm y^{-1}. Experience shows that this rate rapidly decreases with time. Thus strontium release rates were reduced by a factor of 10^6 in 15 y. Figure 22.13 shows measurements of the fraction of Cs released from spent fuel as a function of contact time with simulated ground water. By the time, the fuel matrix will be converted into hydrous oxide. The fraction altered has been measured under various conditions and are extrapolated to very long times in Figure 22.14. These results indicate that even if the barrier surrounding the canisters break down, the leakage of waste products into the near field would be quite small.

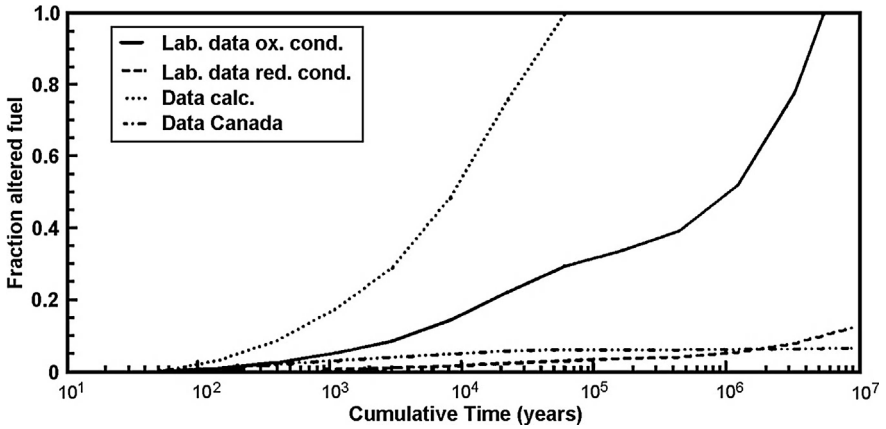


Figure 22.14 Fraction of fuel altered in ground water as a function of time; the reaction is assumed to start 40 years after fuel discharge from the reactor (*From SKB91*).

Table 22.10 Radionuclide solubilities and limiting phase in reducing and oxidizing Finnsjö fresh water (SKB91)

	Reducing conditions		Oxidizing conditions	
	Solubility (M)	**Limiting phase**	**Solubility (M)**	**Limiting phase**
Se	very low	M_xSe_y	high	–
Sr	1×10^{-3}	Strontianite	1×10^{-3}	Strontianite
Zr	2×10^{-11}	ZrO_2	2×10^{-11}	ZrO_2
Tc	2×10^{-8}	TcO_2	high	–
Pd	2×10^{-6}	$Pd(OH)_2$	2×10^{-6}	$Pd(OH)_2$
Sn	3×10^{-8}	SnO_2	3×10^{-8}	SnO_2
I	high	–	high	–
Cs	high	–	high	–
Sm	2×10^{-4}	$Sm_2(CO_3)_3$	2×10^{-4}	$Sm_2(CO_3)_3$
Am	2×10^{-8}	$AmOHCO_3$	2×10^{-8}	$AmOHCO_3$
Pu	2×10^{-8}	$Pu(OH)_4$	3×10^{-9}	$Pu(OH)_4$
Pa	4×10^{-7}	Pa_2O_5	4×10^{-7}	Pa_2O_5

The dissolution rate of canister encapsulation and waste matrix is limited by water solubility. As mentioned above, of the actinides the most hazardous, plutonium, would most likely be in the +4 state and form a very insoluble hydroxide. This is particulary true in an H_2 andFe(II) dominated environment caused by dissolution of an iron canister. Table 22.10 gives limiting solubilities and phases for the more important waste elements in reducing and oxidizing groundwater. Figure 22.15 shows the leakage rate from the near field of an initially defective canister, extrapolated to long times; all nuclides that are expected to leak out at a rate of more than 1 Bq y^{-1} are included in the Figure. In this prediction, the stipulations mentioned in the beginning of §22.10 are met.

22.10.3. Releases from bitumen and concrete encapsulations

The low-level and intermediate-level wastes are to be encapsulated in bitumen, concrete or glass. The bitumen would be highly water resistant, but it ages with time and begins to lose strength in 10–20 years. Special bitumen materials have to be developed for wastes which must be contained in >50 years. The bitumen drums are normally stored in a containment building, e.g. of concrete.

The groundwater surrounding concrete rapidly becomes very basic, pH>13, as a consequence of leakage of Na^+ and K^+ ions. At a later stage, Ca^{2+} ions begin to leak, reducing the pH to about 10.5. Thus nuclides migrating through the concrete encounter very basic media, leading to the formation of insoluble hydroxides for most polyvalent ions, including all actinides. The leaching of Ca^{2+} from the concrete causes it to lose its mechanical strength and to begin to deteriorate. By suitable additives, the onset of deterioration can be delayed, and at pH<10 the concrete is stable for very long times.

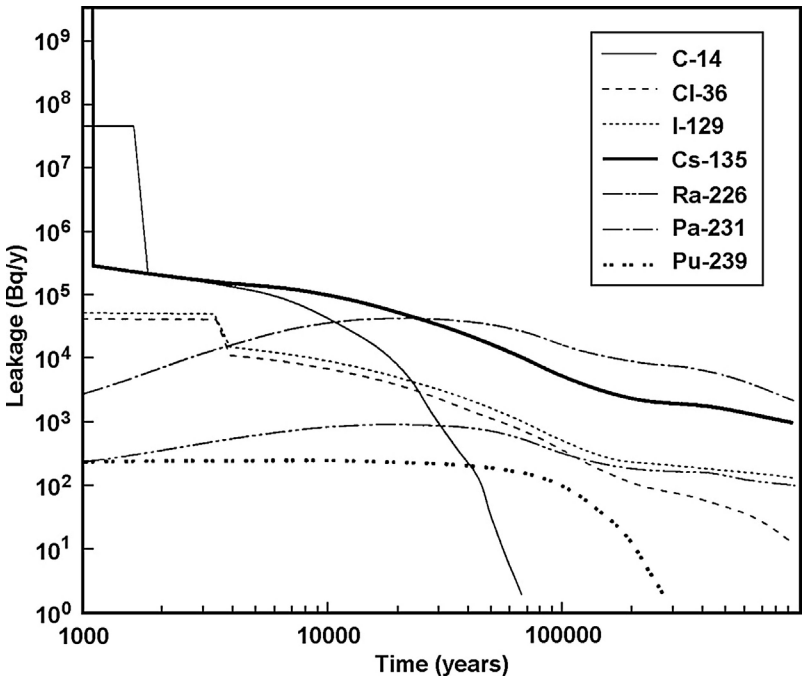


Figure 22.15 Leakage from the near field of radionuclides from an initially defect canister; all nuclides that leak out at a rate of >1 Bq/y are included (*From SKB91*).

However, concrete containers for LLW and MLW can not be considered to have an infinite lifetime, and therefore further containment is necessary.

22.10.4. Migration from the repository

After release into the near field, the radionuclides can only migrate via a water transport path (Figure 22.12). Migration in the far field may occur for radionuclides with long lifetimes, high solubility in ground water, and low sorption along the transport pathway. In repositories where the water is confined to interstitial fracture and pore areas, the actinides would have low solubility and high sorption compared with the fission products. All nuclides which adsorb would move slower than the free groundwater due to the sorption-desorption equilibria. As a consequence ^{129}I and ^{99}Tc would be expected to be the principle contributors to a radioactivity release to the environment, as they are poorly adsorbed. This is also true for ^{14}C, which contributes because it is transported as dissolved CO_2 (Fig. 22.15).

The effect of elevated temperatures resulting from the energy released by the radioactive decay must be included in the evaluation of release and near field migration (i.e. within the repository volume). Elevated temperatures could alter the geology as well

as the chemical speciation and solubility of the released nuclides. If the temperature exceeds the boiling point of water in the fluids, it would result in a drier repository with reduced or no release and migration.

When granite or clay is contacted with water containing dissolved cations, sorption or exchange of these ions with ions of the solid phase are observed. For example, montmorillonite has such a high exchange capacity that it is used as a natural ion exchanger, e.g. for water purification. The ion exchange elution curves in Figure 14.9, however, do not depend on sorption capacity, but on sorption strength and on aqueous complexation. The sorption strength depends on ionic charge (higher charged species sorb more strongly), ionic size (smaller ions sorb more strongly), etc., while aqueous complexation depends on the nature of the complexant (ligand), as well as on the cation properties.

All ground/soil/rock materials sorb ions. This sorption has traditionally been defined as the sorption distribution, k_d, eqn. (22.12). The k_d depends on so many factors that they may be considered site specific; i.e. dependent on the ion released, the near field matrix composition (buffer material, dissolved canister, etc), the ground water composition and the rock mineral composition. However, clay and granite have about the same k_d values in similar groundwaters. A more recent redefinition claim that instead use the term k_d only for one species while the complete sorption is described by the term R_d analogous to D and k_d in solvent extraction. However, most of the collected literature data use the older definition why we for simplicity also do so.

$$k_d = \frac{\text{conc. of radionuclide per kg soil/rock/etc}}{\text{conc. of radionuclide per m}^3 \text{ water}} \tag{22.12}$$

Typical k_d values are listed in Table 22.11. From the k_d values and the soil and groundwater properties, the retention time for a radionuclide may be calculated. The radionuclide retention factor (RF) is defined as

$$RF = v_w/v_n = 1 + k_d\,\delta(1-\varepsilon)/\varepsilon \tag{22.13}$$

Table 22.11 Far field radionuclide distribution values for granitic rock (SKB91)

Element	k_d value ($m^3\ kg^{-1}$)	Element	k_d value ($m^3\ kg^{-1}$)
Zirconium	1	Carbon	0.001
Radium	0.15	Chlorine	0
Protactinium	1	Palladium	0.001
Thorium	2	Selenium	0.001
Uranium	2	Strontium	0.015
Neptunium	2	Cesium	0.15
Plutonium	0.2	Iodine	0
Technetium	1		

Conditions: specific surface area 0.1 m^2 per m^3 of rock, (equal to) 100 m^2 per m^3 of water; matrix diffusion coefficient 3.2×10^{-6} $m^2\ y^{-1}$; diffusion porosity in the rock matrix 0.005.

where v_w is the groundwater velocity (in granite typically 0.1 m y^{-1}), v_n the nuclide transport velocity (which must be $<v_w$), δ the soil density (typically 1 500−2 500 kg m^{-3}), ε the soil porosity (or void fraction, typically 0.01−0.05), The groundwater velocity can be obtained from eqn. (21.15), if geologic parameters are known. Observed k_d values for Swedish granite are given in Table 22.11. Typical retention times in granite are about 400 y for Cs, 1 500 y for Sr, 200 000 y for Ln(III) and 40 000 y for An(IV). Rather similar (within a factor of ten) values have been found for other geologic formations such as tuff from New Mexico, basalt from Idaho, and limestone from Illinois. The high retention values combined with a water transport time of a few years lead to retention of ^{90}Sr, ^{137}Cs, and most Pu, Am, and Cm isotopes over their lifetimes. With the negligible water flow rate through clay, diffusion is the dominating transport process. It is estimated that it will take 700 y for Cs, 1 800 y for Sr, 22 000 y for Am and 8 000 y for Pu to penetrate 0.4 m bentonite. During those times most of the radioactivity of these nuclides have decreased to negligible values. Studies for a salt repository (WIPP) in New Mexico indicate that, of all waste nuclides released in the repository, only ^{14}C, ^{99}Tc, and ^{129}I may reach the surface before they have decayed. Thus, ground retention of the radionuclides plays an essential role in the risk evaluation.

Both dissolution rate and transport rate of nuclides depends on the existence of complex formers in the groundwater. Such complex formers are Cl^-, F^-, SO_2^{4-}, HPO_4^{3-}, CO_3^{2-}, and organic anions (e.g. humic acid). Complexes with these anions, in most cases, increase solubility, and − through formation of less positively charged metal species − reduce the retention factors. The groundwater conditions, therefore, play a central role in evaluation of the risks of a waste repository. Since these conditions vary, they have to be evaluated for each site. This point is illustrated by the Maxey Flats facility in the USA, where radionuclides were found to move from a storage basin much more rapidly through soil than would be expected from the equations above. This was found to be caused by the addition of strong metal chelating complex formers like diethylene-triaminopentaacetic acid (DTPA) to the basin liquid. The DTPA complexes have negative charge, resulting in lower k_d values (usually $\sim$1). This reflects the necessity to know in adequate detail the chemistry of the waste and of the storage site for proper evaluation of the safety.

Figure 22.2 shows the result of a safety analysis, indicating the estimated amounts of released radioactivities carried by ground water from a spent fuel repository to a recipient, and the expected doses delivered, without consideration of retention. Retention will cause the ^{226}Ra and ^{231}Pa to reach zero value before 10 000 years.

Calculations for individual doses and the risks for intrusion into a granite repository of LWR spent fuel at 100 and at 100 000 years after burial are listed in Table 22.12. These calculations are based on very extensive modelling (cf. Fig. 22.12) of the releases and uptakes from wastes produced from a 20 GWe reactor park operating in Europe for

Table 22.12 Individual doses and risks from intrusion scenarios in a granite repository of unreprocessed LWR spent fuel (Mobbs et al.)

	100 years		100 000 years	
Radionuclide	**Dose Rate (Sv y^{-1})**	**Risk[†] (y^{-1})**	**Dose Rate (Sv y^{-1})**	**Risk[‡] (y^{-1})**
^{99}Tc	5.0×10^{-6}	1.5×10^{-14}	1.9×10^{-9}	2.5×10^{-14}
^{129}I	1.8×10^{-6}	5.4×10^{-16}	3.2×10^{-10}	4.2×10^{-15}
^{135}Cs	2.4×10^{-6}	7.0×10^{-15}	1.3×10^{-7}	1.7×10^{-12}
^{137}Cs	1.5	1.8×10^{-7}	—	—
^{234}U	7.3×10^{-4}	2.1×10^{-12}	5.0×10^{-7}	6.5×10^{-11}
^{235}U	1.8×10^{-5}	5.1×10^{-14}	1.4×10^{-6}	1.9×10^{-11}
^{236}U	1.9×10^{-4}	5.4×10^{-13}	2.8×10^{-7}	3.7×10^{-12}
^{238}U	2.0×10^{-4}	5.9×10^{-13}	5.9×10^{-7}	7.7×10^{-12}
^{237}Np	1.1×10^{-3}	3.1×10^{-12}	3.6×10^{-6}	4.7×10^{-11}
^{238}Pu	3.6	1.8×10^{-7}	5.1×10^{-6}	6.7×10^{-11}
^{239}Pu	1.0	1.8×10^{-7}	1.2×10^{-4}	1.5×10^{-9}
^{240}Pu	1.5	1.8×10^{-7}	3.1×10^{-7}	4.0×10^{-12}
^{241}Pu	1.0×10^{1}	1.8×10^{-7}	9.2×10^{-6}	1.2×10^{-9}
^{242}Pu	5.4×10^{-3}	1.6×10^{-11}	8.9×10^{-6}	1.2×10^{-10}
^{241}Am	2.0	1.6×10^{-7}	1.8×10^{-6}	2.4×10^{-11}
^{243}Am	6.4×10^{-2}	1.9×10^{-10}	3.4×10^{-6}	4.4×10^{-11}
^{244}Cm	8.7×10^{-2}	2.5×10^{-10}	4.7×10^{-9}	6.2×10^{-14}
^{245}Cm	4.8×10^{-4}	1.4×10^{-12}	7.3×10^{-9}	9.6×10^{-14}
Total	2.1×10^{1}	1.8×10^{-7}	1.6×10^{-4}	2.1×10^{-9}

[†]From S. F. Mobbs et al

[‡]Dose-risk conversion factor = 1.65×10^{-2} Sv^{-1} recommended by ICRP in 1977 [ICRP, 1977]. However, ICRP recommended a higher value, 5.0×10^{-2} Sv^{-1}, in 1990 [ICRP #60, 1990].

30 years, including reprocessing and storage of LLW and ILW in surface facilities, and HLW, either in the form of spent fuel elements or vitrified HLLW in an underground granite repository. To model the dissolution of radionuclides and their migration in the ground existing facilities in their actual environments have been chosen. Similarly, for the uptake actual food chains (crops, fish , etc and eating habits) have been used. The quantity of spent fuel is assumed to be 18 000 t IHM. The LLW and ILW repositories are assumed to begin to leak at time 0, the dissolved radionuclides migrating into the actual soils, reaching streams and wells, etc, while it is assumed that the spent fuel elements or high level waste glass begins to leak after 1 000 years. The radionuclides migrate by groundwater under various retention conditions, depending on hydrogeologic conditions and chemical properties. The exposed population is divided into four groups: the critical group around the installations, the population of the country, the population in Europe and in the world. As expected, the population dose is the highest for the critical and national groups for aqueous releases, but about the same for all groups for gaseous

releases. The model also includes an intrusion scenario using a well drilling probability, beginning after 300 years (in the main case), as it is expected that control will last that long. It is assumed that as soon as a drill core from a waste repository is taken, it will immediately be recognized; this means that high doses from the core will only be received by drilling and laboratory personal. The risk[1] from intrusion after 100 years is calculated to be 8.1×10^{-7} y^{-1} and after 100 000 years 2.1×10^{-9} y^{-1}. By comparison, the calculated risk from the "normal" scenario of releases from spent fuel elements and migration in the environment over a long period is 5×10^{-5} y^{-1}. With this exception all the maximum calculated individual risks from disposal of solid wastes are below the limit of 10^{-5} y^{-1} recommended by the ICRP. For vitrified high level waste and spent fuel disposal, the doses calculated for intrusion scenarios are very high but the probability of occurrence are low, so the risks from intrusion are lower than those from radionuclide migration with groundwater. The maximum individual doses from the migration scenarios for solid waste disposal are predicted to arise beyond 10 000 years after closure of the repository.

These values indicate that release and migration is the more significant concern, although even for this scenario, geologic disposal is calculated to present very small risks to future generations.

22.11. CONCLUSIONS

A rather large amount of nuclear fission products and actinide elements have been released to the environment from nuclear weapons testing and from accidental and intentional discharges from nuclear reactor operations and fuel reprocessing. The research on the fate of these released radionuclides suggest that the long lived actinides form quite insoluble or strongly sorbed species while ^{129}I and ^{99}Tc have relatively high dissemination in natural systems. The most active shorter-lived species (^{90}Sr, ^{137}Cs) also have more mobility in ecosystems.

These conclusions are confirmed by the results of the investigations of radionuclide behavior in natural analog sites. Even more relevant are the results of the natural reactor region at Oklo.

The data from field studies is largely confirmed by the performance assessments of proposed nuclear waste repositories. The "intrusion" scenario is calculated to have a lower risk than the undisturbed natural leach and migration scenario. The latter qualitatively agrees with the Oklo data and indicates no unacceptable risks result from a carefully chosen and designed geologic repository in which the nuclear wastes are emplaced with appropriate packaging.

[1] Risk is here assumed to be equal to expected frequency regardless of the severity of the consequence.